Emergent Born Rule via Kinematically Constrained Quantum Scrambling

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Abstract

A novel derivation of the Born rule is provided within a highly scrambled many-body environment. Randomness in quantum probability is significantly reduced to classical probability on the Haar measure. Furthermore, the order in which the Born rule is derived from envariance arguments for general unequal amplitude cases is reversed. Instead of fine-graining states with unequal amplitudes to states with equal amplitudes, fine-grained states with effectively equal amplitudes that arise from Haar-random scrambling are coarse-grained as to produce the Born rule. The result, proved largely as a consequence of Page's theorem and the envariance argument in the thermodynamic limit, arises from treating the late-time scrambled sector as a Haar-random subspace, motivated by the eigenstate thermalization hypothesis and non-perturbative quantum gravity. Explicitly deriving the Haar-random ansatz from first principles is left to future works.

1 Introduction

Two of the most profound conceptual tensions in theoretical physics concern the foundational origin of quantum probability and the role of ensemble randomness in quantum chaos. In standard quantum mechanics, the Born rule - the prescription that a measurement outcome occurs with probability $p_k = |\psi_k|^2$ - is introduced as an irreducible axiom with no dynamical derivation [1]. Meanwhile, explaining the universal chaotic signatures of the Spectral Form Factor (SFF) in many-body systems has traditionally relied on drawing Hamiltonians from random matrix ensembles [2, 3]. This statistical averaging procedure introduces fundamental randomness that appears to conflict with the strict unitary evolution mandated for isolated quantum systems.

We demonstrate that these problems admit a common partial resolution within a single kinematic framework, though the emphasis is placed on the Born rule in this work. The logical structure of this work is provided in Figure 1. The central postulate is that the fundamental Hamiltonian of a complex system is strictly fixed and non-random. The appearance of ensemble randomness in the SFF rather emerges as a kinematic consequence of projecting unitary evolution onto a Haar-random subspace of late-time, fully scrambled states - a subspace whose geometry is set by the specific boundary conditions of the universe instance we inhabit. The kinematic Haar-random ansatz is motivated by the Eigenstate Thermalization Hypothesis and by holographic (quantum) gravity considerations, but is not derived from first principles within this work - the explicit 'from first principles' derivation is left to future works.

Since we are not developing a realistic model of quantum, we restrict our framework to involve a strongly chaotic non-integrable many-body systems satisfying Eigenstate Thermalization Hypothesis.

1.1 Main Objectives

We emphasize that the fundamental part of this paper is connecting the kinematic Haar-random ansatz with Page's theorem and the envariance derivation of the Born rule, not the SFF results. The main objectives are as follows:

1. Providing the kinematic Haar-random justification of the envariance derivation of the Born rule. That is, we reduce large parts of randomness in quantum probability to classical probability (over the Haar measure on α).

2. Evading the unequal amplitude complications of the envariance argument by modeling measurements as Haar-random scrambling. Even if the initial actual state of S has unequal amplitudes on the measurement basis/observable $\hat{O}$, the final fine-grained ensemble state has equal amplitudes, and the Born rule arises from coarse-graining. Typical envariance arguments go in the other direction - we have to 'fine-grain' states with unequal amplitudes to equal-amplitude states, which have been considered problematic [4, 5].

We also note that this work does not aim at providing a realistic model of quantum measurements, only that the justification for the Born rule is logically much stronger within unitary quantum mechanics than previously thought.

1.2 Secondary Objective: On the Role of the SFF Results and Replica Wormhole Purity

The dip-ramp-plateau structure follows from a constrained, though not too restrictive, set of conditions, and we emphasize that the SFF structure is a conditional consequence, not a general result. Within JT gravity, this result is already widely known [6, 7], and the JT gravity replica wormhole result is discussed in Section 3.6.

The SFF results are not a fundamental component of the Born rule derivation and are largely restatements of existing results (with randomness moved from ensemble theory to subspace, especially within JT gravity). If we assume some mechanism of wavefunction collapse or the Everettian interpretation, SFF results are unnecessary, and the Born rule derivation follows directly from the kinematic ansatz under more general conditions. However, if we want unitary deterministic quantum mechanics, it must be established that the actual (per-instance) state of system S is approximately pure. This purity question is resolved in the present work by Proposition 3.11, which uses the replica trick and the gravitational path integral over competing saddle geometries - disconnected disk contributions vs. connected wormhole topologies - to show that the n-th Rényi entropy of ρ_S satisfies $\mathrm{Tr}(\rho_S^n) \to 1$ at late times, hence $S(\rho_S) \to 0$. Despite the name 'gravitational path integral,' there is not really much about gravity and is mostly about integrating over geometries that suggest how different replicas of the same system are to be connected. This method has been controversial within non-ensemble contexts, but much controversy can be avoided because we are working with ensembles.

As we will see in Corollary 4.4, the final fine-grained ensemble state of system S is approximately maximally mixed. (Equivalently, the final fine-grained ensemble state of the universe is heavily entangled.) Once Proposition 3.11 establishes that the actual state of S is approximately pure, any basis can be used to derive the Born rule via coarse-graining - ignorance of the actual basis is not an issue, as long as interaction preserves amplitude information on the measurement basis, which can be chosen as pointer basis with observable $\hat{O}$ $[H_{\mathrm{int}}, \hat{O}] = 0$. This self-contained wormhole argument substitutes any appeal to wavefunction collapse or many-worlds interpretation for the purpose of establishing purity. However, even without the SFF results, as long as purity of S can be granted, any basis can be used to derive the Born rule, since the fine-grained ensemble state of system S is approximately maximally mixed.

1.3 Scope and Assumptions

1.3.1 The Postulates and Axioms

The framework rests on four explicit postulates (along with history-independent state-dependent probability and additive probability for disjoint events, which are already standard assumptions):

1. **Unitary evolution**: The total system evolves unitarily under a fixed Hamiltonian.

   (The total system evolves unitarily under $H = H_{\text{probe}} + H_E(\alpha) + \lambda H_{\text{int}}$.)

2. **No-signaling**: Local operations on a subsystem cannot alter the statistics of a causally disconnected subsystem - relativistic causality.

   (Local operations on environment E cannot alter the statistics of S)

3. **Effective Haar-random ansatz**: The late-time scrambled sector is well-described by a Haar-random subspace of the full Hilbert space (asymptotically justified by ETH - see Remark 3.2).

   (The late-time scrambled sector $\mathcal{V}_\alpha$ is described by a Haar-random isometry Φ_α (Remark 3.2). This is an effective late-time description, not a fundamental postulate.)

4. **Continuity**: The probability map is continuous in the ℓ^2 norm on state space (a regularity assumption required only for irrational amplitudes - see Assumption 6.1).

   (The probability map is continuous in the ℓ^2 norm (required only for irrational amplitudes; structurally analogous to Gleason's measurability condition).)

5. **State-dependence, history-independence**: Outcome probabilities depend only on the state, not on history.

6. **Additivity**: Probabilities of disjoint events are additive.

1.3.2 What are derived

The following is what are derived:

- The SFF ramp-plateau structure (Theorem 3.6, Eq. (13), largely restatements)
- The structural origin of the dip as a disconnected/connected crossover (Proposition 3.8, Remark 3.9, largely restatements),

- Per-instance (per-α) spectral resolution from the SFF plateau (Remark 3.10) (largely restatements),
- Late-time purity of ρ_S from the replica trick and wormhole saddle dominance (Proposition 3.11, largely restatements),
- Exact branch preservation from the tensor product structure independently of the Haar measure (Proposition 4.1)
- The exponential closeness of the local reduced density matrix to the maximally mixed state (Corollary 4.4, Lemma 4.5)
- Exact equiprobability for fine-grained branches in any measurement basis without a preselected pointer basis (Theorem 5.2, Remark 5.4) - therefore, decoherence-based pointer basis and the preferred basis problem are separate from the Born rule derivation in this work. However, this assumes that interaction preserves amplitude preserves amplitude information, and imposing this requires $[H_{\text{int}}, \hat{O}] = 0$, with measurement observable being $\hat{O}$.
- The Born rule for all amplitudes (Section 6), and the interpretation of the Born rule as a classical probability over the Haar measure on the scrambled-sector microstate α, with each instance α yielding a deterministic, pure-state outcome for S, if we adopt the replica wormhole result or some quantum interpretational resolutions.

1.3.3 What are not derived and not solved

The following lists what are not derived and not solved:

- The postulates and axioms,
- The precise depth and location of the dip minimum (which require specifying H_{base}'s spectrum, such as JT gravity),
- The measurement problem and a realistic model of quantum measurements,
- The explicit derivation of the kinematic Haar-random ansatz from the first principles.

1.4 Organization

Section 2 provides physical intuition through a holographic brane/non-brane analogy and a qubit toy model of black hole evaporation, motivating the kinematic framework and clarifying its relationship to ensemble-based approaches in JT gravity. Section 3 defines the kinematic constraint formally and derives the SFF statistics, including a treatment of the disconnected SFF and the dip in Section 3.5, a remark (Remark 3.10) on per-instance spectral resolution, and a new Section 3.6 establishing late-time purity via the replica trick and wormhole saddles. Section 4 applies Page's theorem to establish maximal local mixing, including Proposition 4.1 on exact branch preservation. Section 5 derives equal probabilities from envariance and no-signaling in the thermodynamic limit. Section 6 extends the result to the general Born rule. The roadmap for this work is provided in Figure 1.

2 Physical Intuition and Holographic Motivation

Before stating our framework, we provide some physical intuition. The abstract structure - a Hilbert space split into a pre-scrambling sector governed by H_{base} and a post-scrambling sector governed by H_{post}, joined by a Haar-random projector - has a natural interpretation in the language of holography and black hole physics. We develop this interpretation in three stages: a brane/non-brane decomposition, a simplified qubit toy model that makes the mechanism concrete, and a discussion of the relationship to ensemble-based approaches in JT gravity that motivates our specific "subspace" rectification.

2.1 The Brane/Non-Brane Decomposition

In holographic settings, a useful organizing principle is the separation between *non-brane* (bulk) degrees of freedom and *brane* (boundary or end-of-the-world) degrees of freedom. In our framework, this separation maps onto the Hamiltonian structure as follows:

- H_{base} governs the *non-brane sector*: the early-time, pre-scrambling dynamics of the local probe and the adjacent environment. In holographic language, this corresponds to the black hole exterior - the region accessible to an asymptotic observer before significant information transfer has occurred.
- H_{post} governs the *brane sector*: the late-time, fully scrambled degrees of freedom. In

holographic language, this corresponds to the black hole interior or, more precisely, to the effective degrees of freedom of the end-of-the-world (ETW) brane that emerges in certain holographic models of evaporation [6, 9, 7].

- The Haar-random projector Π_α defines the interface between these two sectors. It encodes the specific microstate - the particular scrambled configuration - of the universe instance α.

Note that we do *not* claim that the framework requires a black hole or a holographic dual. The brane/non-brane language is a motivating analogy. The formal content of the framework is entirely within standard quantum mechanics: a fixed Hilbert space, a fixed Hamiltonian, and a kinematic constraint on the late-time sector. The holographic language is intuition and a potential source of evidence for the Haar-random ansatz.

2.2 A Simplified Qubit Toy Model of Black Hole Evaporation

To make the transition from H_{base} to H_{post} concrete, we construct a minimal toy model. This model is not a realistic description of Hawking radiation, but it provides intuition for why the late-time sector requires a *different* effective Hamiltonian.

State space. Consider a total Hilbert space

$$\mathcal{H}_{\text{toy}} = \mathcal{H}_{\text{int}} \otimes \bigotimes_{k=1}^{M} \mathcal{H}_k^{(q)}, \tag{1}$$

where $\mathcal{H}_{\text{int}}$ is a $(M+1)$-dimensional interior Hilbert space spanned by energy eigenstates $|0\rangle_{\text{int}}, |1\rangle_{\text{int}}, \ldots, |M\rangle_{\text{int}}$ (the ground state plus M excited levels), and each $\mathcal{H}_k^{(q)} \cong \mathbb{C}^2$ is a qubit representing a single exterior radiation mode.

Initial state. The interior begins in its maximally excited state $|M\rangle_{\text{int}}$, and all exterior qubits begin in their ground state:

$$|\Psi_0\rangle = |M\rangle_{\text{int}} \otimes |0\rangle_1 \otimes |0\rangle_2 \otimes \cdots \otimes |0\rangle_M. \tag{2}$$

This initial state represents a black hole of "mass" M in energy units, with no Hawking radiation yet emitted.

Evaporation dynamics. The early-time Hamiltonian H_{base} drives the following sequential process. At step k, the interaction

$$U_k = \exp\left[-i\theta_k \left(a_{\text{int}}^\dagger \sigma_k^- + a_{\text{int}} \sigma_k^+\right)\right], \tag{3}$$

where a_{int} is the interior lowering operator and $\sigma_k^\pm$ are raising/ lowering operators for qubit k, transfers one quantum of energy from the interior to qubit k. Choosing $\theta_k = \pi/2$ gives a complete energy transfer:

$$U_k \left(|n\rangle_{\text{int}} \otimes |0\rangle_k\right) = |n-1\rangle_{\text{int}} \otimes |1\rangle_k, \quad n \geq 1. \tag{4}$$

After step k, qubit k "flies away from the horizon": it becomes causally disconnected from the interior and from all subsequent qubits. The total state after all M steps is:

$$|\Psi_M\rangle = |0\rangle_{\text{int}} \otimes |1\rangle_1 \otimes |1\rangle_2 \otimes \cdots \otimes |1\rangle_M. \tag{5}$$

The interior has reached its ground state $|0\rangle_{\text{int}}$, and all M units of energy have been transferred to the radiation modes.

Why H_{post} is necessary. At $|\Psi_M\rangle$, a critical issue arises. The Hamiltonian H_{base}, which drove the evaporation, was designed to couple the interior to the exterior qubits. Continued action of H_{base} on the final state $|\Psi_M\rangle$ would attempt to lower the interior below its ground state, which is unphysical. More concretely, $a_{\text{int}}|0\rangle_{\text{int}} = 0$, so the drive term $a_{\text{int}}\sigma_k^+$ acting on $|\Psi_M\rangle$ yields zero - but the Hermitian conjugate term $a_{\text{int}}^\dagger \sigma_k^-$ would attempt to re-absorb radiation from qubits that either have no energy to emit, are non-existent or are already de-coupled, violating causality. The base Hamiltonian simply becomes inappropriate once the ground state is reached.

The role of H_{post} is precisely to govern the late-time dynamics in this post-evaporation regime. It describes the physics of the interior at and near its ground state under the constraint that the evaporation process has concluded. In the formal framework of Section 3, H_{post} governs the N-dimensional scrambled subspace $\mathcal{V}_\alpha$ that the interior has settled into after full evaporation. Its spectrum $\{E_n\}$ encodes the fine-grained structure of the post-evaporation ground manifold, and its Wigner-Dyson statistics follow from the chaotic nature of the scrambling process that

brought the system there.

In terms of the total Hamiltonian (see Eq. (7) in Section 3), the transition from H_{base}-dominated to H_{post}-dominated dynamics is not a theory (or dynamics) discontinuity but the asymptotic consequence of the interior exhausting its excitations. We emphasize, though, this is only an analogy and should only be used for building intuition.

2.3 JT Gravity, Ensembles, and the Born Rule Question

The Jackiw-Teitelboim (JT) gravity model [6] provides the most developed field-theoretic framework in which black hole information, Haar-random scrambling, and the spectral form factor can all be discussed rigorously. We briefly describe the relevant structure and identify the question that our framework is specifically designed to address.

JT gravity as a matrix integral. JT gravity in two dimensions is 'equivalent', as a path integral over geometries, to double-scaled random matrix theory: a statistical ensemble of $(L \times L)$ Hamiltonians drawn from a random matrix distribution in the double-scaling limit (which includes the $L \to \infty$ limit) [6]. The gravitational path integral computes ensemble averages of boundary observables. In this framework:

- The SFF, computed as a sum over spacetime topologies including wormholes connecting different replicas, reproduces the dip-ramp-plateau structure characteristic of Wigner-Dyson universality.
- The black hole information paradox is 'resolved', at the ensemble level, by the contribution of replica wormhole topologies to the entanglement entropy. The resulting Page curve [9] matches what is expected of a unitary evaporation process.
- The relevant scrambling, i.e., the mapping of infalling information into the outgoing Hawking radiation, is accurately modeled by a Haar-random unitary on the full Hilbert space of the radiation, consistent with the Hayden-Preskill framework [10].

The replica wormhole origin of the dip. The three-part dip-ramp-plateau structure of the SFF in JT gravity arises from a specific topological decomposition of the gravitational path integral. Computing $K(t) = |\mathrm{Tr}\, e^{-iHt}|^2$ requires summing over geometries connecting two boundary circles. The dominant topologies are:

- *Disconnected saddle* (two separate disks): contributes $K_{\text{disconn}}(t) \sim |\hat{\rho}_{\text{Sch}}(t)|^2$, where ρ_{Sch} is the Schwarzian density of states $\rho_{\text{Sch}}(E) \propto e^{S_0} \sinh(2\pi\sqrt{E})$. This decays at early times by the Riemann-Lebesgue lemma applied to the smooth spectral density.

- *Connected saddle* (double-trumpet wormhole): contributes $K_{\text{conn}}(t) \sim t/(2\pi)$, the linear ramp.

Both topologies are *mandatory* in the gravitational path integral: one cannot exclude either saddle. The dip minimum is therefore topologically guaranteed - it is the inevitable crossing of the decaying disconnected piece and the growing ramp. Its location t_{Th} and depth are set by the specific Schwarzian spectral density.

This has a direct counterpart in our kinematic framework. The H_{base} sector plays the role of the disconnected disk contributions; the H_{post} sector restricted to $\mathcal{V}_\alpha$ plays the role of the wormhole contribution (and provides a source of ensemble randomness that justifies connected contributions).

Question A: What justifies the Haar-random distribution over theories? The ensemble in JT gravity consists of Haar-random random matrix Hamiltonians: the "theories" themselves are drawn from a Haar-distributed ensemble of unitary matrices. But this raises a serious physical question. If we live in a universe described by a single Hamiltonian, what justifies treating our theory as a typical draw from an ensemble of Haar-random theories? The ensemble interpretation seems to require either:

- A many-worlds reading in which all ensemble members are simultaneously realized (which shifts the foundational burden to the interpretation of the ensemble weight), or

- A subjective Bayesian reading in which the ensemble represents our ignorance about which theory describes our universe (which introduces prior probabilities that are themselves in need of justification).

Neither option is fully satisfactory from a foundational standpoint.

2.4 From Theory Ensemble Randomness to Subspace Randomness: The Rectification

Our framework is designed to provide a specific rectification of Question A.

The key shift: subspace over theory ensemble. Instead of distributing randomness over *theories* (different Hamiltonians), we distribute it over *subspaces* of a single fixed theory. The Haar-random unitary $V_\alpha \in G$ does not define a random Hamiltonian; it defines a random N-dimensional subspace $\mathcal{V}_\alpha$ of the fixed Hilbert space $\mathcal{H}$, within which the fixed, deterministic Hamiltonian H_{post} operates. The physics in any given universe instance α is governed by this single fixed theory; what varies between instances is which subspace the late-time scrambled sector happens to occupy.

This is a physically meaningful distinction:

- In the typical ensemble interpretation of JT gravity, different members have genuinely different Hamiltonians and genuinely different spectra, sharing the same semiclassical bulk theory. The ensemble is a distribution over the space of theories.
- In our framework, the Hamiltonian is fixed and the spectrum of H_{post} is fixed. The only instance-dependent quantity is the orientation V_α of the scrambled subspace within the full Hilbert space.

The Born rule probabilities then arise not from averaging over theories but from the specific geometry of the Haar-random subspace that the system has settled into. In the language of the toy model: the "which qubit carried away which excitation" data - which would specify the exact microstate of the radiation - is unobservable, and its uniform distribution over $\mathcal{V}_\alpha$ is what the Haar measure encodes.

Why a brane/non-brane separation is still required. The subspace mechanism requires a meaningful separation between the sector governed by H_{base} (the non-brane, or early-time sector) and the sector governed by H_{post} (the brane, or late-time scrambled sector). Without this separation, the Haar-random projector Π_α would have no natural embedding: it would be unclear what the "unscrambled" reference sector is relative to which the scrambled subspace is defined.

This is analogous to the role played by the ETW brane in holographic models of black hole evaporation: the brane separates the black hole interior (high-entropy, chaotic, described by a Haar-random state) from the exterior (low-entropy, structured, described by a semiclassical geometry). In our framework, H_{base} plays the role of the exterior Hamiltonian and H_{post} plays the role of the brane dynamics, with Π_α defining the interface.

Summary of the intuitive picture. The physical picture assembled from this section is:

1. A quantum system begins in a low-entropy, structured regime governed by H_{base} (non-brane/exterior sector).

2. Through scrambling dynamics - the many-body analog of black hole evaporation - it transitions into a high-entropy, chaotic regime described by H_{post} restricted to a specific Haar-random subspace $\mathcal{V}_\alpha$ (brane/interior sector).

3. Page's theorem ensures that any local subsystem is exponentially close to maximally mixed in this regime. This allows us to invoke the envariance argument for the Born rule in equal amplitude cases.

4. The replica trick, applied to the gravitational path integral summing disconnected and connected (wormhole) saddles, establishes that $S(\rho_S) \to 0$ at late times, confirming per-instance purity (Proposition 3.11). This is justified by relating H_{post} with connected saddles.

5. Envariance, plus no-signaling then forces Born rule probabilities on local coarse-grained measurement outcomes.

The formal implementation of these steps occupies the remainder of the paper.

3 Kinematically Constrained Scrambling and the Spectral Form Factor

3.1 Setup and Kinematic Ansatz

Let $\mathcal{H}$ be the full Hilbert space of an isolated many-body system, with $\dim \mathcal{H} = L$. We work in the thermodynamic regime $L \to \infty$. We postulate two fixed, non-random operators: a pre-scrambling Hamiltonian H_{base} governing early-time deterministic dynamics, and a boundary Hamiltonian H_{post} with a strictly discrete spectrum $\{E_n\}_{n=1}^{L}$ governing late-time equilibrium degrees of freedom.

Definition 3.1 (Kinematic Constraint)**.** The late-time scrambled sector is an N-dimensional subspace $\mathcal{V}_\alpha \subset \mathcal{H}$, with $N \sim e^{S_{\text{th}}}$, defined by the instance-dependent projector

$$\Pi_\alpha = V_\alpha P_0 V_\alpha^\dagger, \tag{6}$$

Without much loss of generality, pick $P_0 = \sum_{E_n \in [E, E+\Delta E]} |E_n\rangle\langle E_n|$, a fixed canonical reference projector of rank N onto a microcanonical energy shell of H_{post} as an example, and $V_\alpha \in G$ is a Haar-random unitary. The first N columns of V_α form the $L \times N$ Haar-random isometry $\Phi_\alpha : \mathbb{C}^N \to \mathbb{C}^L$, satisfying $\Phi_\alpha^\dagger \Phi_\alpha = I_N$ and $\Phi_\alpha \Phi_\alpha^\dagger = \Pi_\alpha$. The symmetry group $G \in \{U(L), O(L), Sp(L)\}$ is selected by the time-reversal symmetry class of H_{post}, corresponding to GUE, GOE, and GSE universality classes respectively.

Remark 3.2 (Status of the Haar-random ansatz)**.** We treat the Haar-random kinematic constraint as an *effective late-time description* rather than a fundamental microscopic postulate. Three levels of justification support this:

(i) **Symmetry.** The Haar measure is the unique unitarily invariant measure on the Grassmannian $\text{Gr}(N, L)$ of N-dimensional subspaces of $\mathcal{H}$. By the maximum entropy principle [11], it is the distribution that maximizes entropy over all unitarily invariant distributions given only the dimensional constraint N.

(ii) **ETH.** For any non-integrable many-body Hamiltonian satisfying the Eigenstate Thermalization Hypothesis [12, 13], matrix elements of local observables in the energy eigenbasis converge in the thermodynamic limit to Gaussian-distributed values with variance $e^{-S_{\text{th}}/2}$, consistent with Haar-random eigenvectors in the microcanonical shell. This is the primary microscopic justification.

(iii) **Holography.** Non-perturbative gravitational contributions (replica wormholes in JT gravity [6, 7]) and the Hayden-Preskill scrambling framework [10] both support Haar-random late-time dynamics. The connection between our subspace mechanism and these holographic results is discussed in Section 2.4.

The total Hamiltonian. We write the total Hamiltonian in the tripartite form:

$$H = H_{\text{probe}} + H_E(\alpha) + \lambda H_{\text{int}}, \tag{7}$$

where $H_{\text{probe}} = (I - \Pi_\alpha) H_{\text{base}} (I - \Pi_\alpha)$ governs the free early-time evolution of the local probe in the 'non-brane' sector (system S), $H_E(\alpha) = \Pi_\alpha H_{\text{post}} \Pi_\alpha$ is the effective environmental Hamiltonian in the scrambled sector (instance-dependent through Π_α, but fixed for a given instance α), and λH_{int} is coupling that mediates transitions between the two sectors.

The coupling H_{int} acts at the boundary between S and envrionment E. Even if we use the Fermi golden rule that is applicable for weak interaction, for $t \gg t_{\text{dec}} \sim \lambda^{-2}$, the 'probe"s amplitude solely in the non-brane sector decays into the exponentially large target space $\mathcal{V}_\alpha$ (dimension $N \sim e^{S_{\text{th}}}$). This decay is effectively irreversible: the Poincaré recurrence time is $t_{\text{rec}} \sim e^{e^{S_{\text{th}}}}$, exponentially exceeding any physically relevant timescale.

Remark 3.3 (Timescale hierarchy)**.** The framework operates across the following hierarchy of timescales:

- $t_{\text{dec}} \sim \lambda^{-2}$: Decoherence/einselection time, during which the pointer basis is selected by H_{int}.
- t_{Th}: Thouless time, at which information is fully scrambled across $\mathcal{V}_\alpha$ and the SFF linear ramp begins. This is also the time at which $K_{\text{disconn}}(t)$ and $K_{\text{conn}}(t)$ are comparable, defining the dip minimum (see Section 3.5).
- $t_{\text{Page}} \sim e^{S_{\text{th}}/2}$: Page time, at which the connected wormhole saddle begins to dominate the replica path integral for $\text{Tr}(\rho_S^n)$, driving the entropy of ρ_S toward zero (see Section 3.6).
- $t_H \sim e^{S_{\text{th}}}$: Heisenberg time, at which spectral discreteness becomes relevant (SFF plateau).
- $t_{\text{rec}} \sim e^{e^{S_{\text{th}}}}$: Poincaré recurrence time; all physically relevant statements hold for $t \ll t_{\text{rec}}$.

The Born rule derivation in Sections 4–6 applies in the regime $t_{\text{Th}} \ll t \ll t_H$; purity of ρ_S is established for $t_{\text{Page}} \ll t \ll t_{\text{rec}}$ (Proposition 3.11).

At late times $t \gg t_{\text{Th}}$, the coupling to the non-brane sector is exponentially suppressed and the effective compressed Hamiltonian governing the scrambled sector is

$$\mathcal{H}_\alpha^{(N)} \equiv \Phi_\alpha^\dagger H_{\text{post}} \Phi_\alpha, \tag{8}$$

an $N \times N$ self-adjoint matrix whose eigenvalues $\{\epsilon_j(\alpha)\}_{j=1}^N$ depend non-trivially on the instance α through Φ_α.

Remark 3.4 (Non-factorization of the matrix exponential)**.** Higher powers of $\mathcal{H}_\alpha^{(N)}$ satisfy $(\mathcal{H}_\alpha^{(N)})^k = \Phi_\alpha^\dagger H_{\text{post}} (P_\alpha H_{\text{post}})^{k-1} \Phi_\alpha$, where $P_\alpha = \Phi_\alpha \Phi_\alpha^\dagger$. Because $P_\alpha \neq I_L$, the matrix exponential does *not* factorize as $e^{-i\mathcal{H}_\alpha^{(N)} t} = \Phi_\alpha^\dagger e^{-iH_{\text{post}} t} \Phi_\alpha$. The spectral statistics of $\mathcal{H}_\alpha^{(N)}$ must therefore be derived via the resolvent.

3.2 Resolvent, Free Compression, and Spectral Statistics

Define the normalized Cauchy (Stieltjes) transform

$$G_\alpha(z) \equiv \frac{1}{N}\mathrm{Tr}_N\left[\left(z\, I_N - \mathcal{H}_\alpha^{(N)}\right)^{-1}\right], \quad z \in \mathbb{C} \setminus \mathbb{R}. \tag{9}$$

The Neumann series expansion of the resolvent reads

$$\left(z - \mathcal{H}_\alpha^{(N)}\right)^{-1} = \frac{1}{z}\sum_{k=0}^{\infty}\frac{1}{z^k}\left(\Phi_\alpha^\dagger H_{\mathrm{post}}\Phi_\alpha\right)^k. \tag{10}$$

Taking the Haar average $\overline{G_\alpha(z)}$ requires evaluating integrals of products of P_α matrix elements, handled by the Weingarten formula of [14] applied to $V_\alpha \in G$. In the large-L limit with $N/L \to c$, the Weingarten sum is dominated by non-crossing pair partitions; crossing contributions are suppressed by $O(L^{-1})$ [15]. Resumming the non-crossing contributions over all orders k yields:

Theorem 3.5 (Free Compression [14, 15])**.** (From free probability theory; the contribution of this work is the verification that its hypotheses are satisfied by the compressed ensemble under the kinematic constraint of Definition 3.1.) *Let H_{post} be a deterministic self-adjoint $L \times L$ matrix whose empirical spectral measure $\mu_H = L^{-1}\sum_n \delta_{E_n}$ converges weakly to a compactly supported probability measure μ as $L \to \infty$. Let Φ_α be a Haar-random isometry with $N/L \to c \in (0,1)$. Then the empirical spectral measure of $\mathcal{H}_\alpha^{(N)}$ converges almost surely to the deterministic measure μ_c whose Cauchy transform $G_c(z)$ satisfies the subordination equation*

$$G_c(z) = \frac{1}{c}\, G_\mu\!\left(z - \frac{1-c}{c}\,\frac{1}{G_c(z)}\right), \tag{11}$$

where $G_\mu(w) = \int (w-\lambda)^{-1} d\mu(\lambda)$ is the Cauchy transform of μ.

The mean spectral density is recovered by Stieltjes inversion: $\rho_c(\epsilon) = -\frac{1}{\pi c}\lim_{\eta\to 0^+}\mathrm{Im}\, G_c(\epsilon + i\eta)$.

3.3 Wigner-Dyson Statistics and the Ramp-Plateau

Theorem 3.6 (Wigner-Dyson Universality [16, 17])**.** (From random matrix universality theory; the contribution of this work is the verification that rotational invariance of the compressed ensemble follows from the Haar measure in Definition 3.1.) *For any fixed W in the N-dimensional symmetry group, $\mathcal{H}_\alpha^{(N)} \overset{d}{=} W^\dagger \mathcal{H}_\alpha^{(N)} W$. By the universality theorem for rotationally invariant*

matrix ensembles, the local bulk eigenvalue statistics of $\mathcal{H}_\alpha^{(N)}$ converge to the GUE (resp. GOE, GSE) sine-kernel statistics for $G = U(L)$ (resp. $O(L)$, $Sp(L)$), independently of the particular spectral measure μ of H_{post}.

Given the sine-kernel two-point correlation in unfolded units $s = N\rho_c(\epsilon_1 - \epsilon_2)$,

$$R_{2,c}(s) = \rho_c^2 \left[1 - \left(\frac{\sin \pi s}{\pi s} \right)^2 \right], \tag{12}$$

the connected SFF follows by Fourier transform of the cluster function [3]:

$$K_{\text{conn}}(t) = \begin{cases} |t|/(2\pi) & |t| \le t_H, \\ t_H/(2\pi) \approx N\rho_c(E_0) & |t| > t_H. \end{cases} \tag{13}$$

This is the ramp-plateau of the connected SFF. The full SFF, including the dip, requires also the disconnected piece; see Section 3.5.

Remark 3.7 (Role of Weingarten calculus)**.** The Weingarten formula of [14] enters at the level of the resolvent expansion (10). The explicit Weingarten functions $\mathrm{Wg}(\sigma\tau^{-1}, L)$ govern the $O(L^{-|\sigma\tau^{-1}|})$ corrections; resummation over non-crossing partitions recovers the subordination equation (11). For $O(L)$, the Weingarten functions are replaced by zonal spherical functions [14], yielding a ramp coefficient twice that of GUE; for $Sp(L)$ the coefficient is halved, reproducing the full Wigner-Dyson universality classification.

3.4 Plateau and Heisenberg Time

The linear ramp of Eq. (13) saturates at the Heisenberg time $t_H = 2\pi N\rho_c$, set by the mean level spacing $\Delta = 1/(N\rho_c)$ of the compressed spectrum. Spectral rigidity [18] constrains the number variance to grow only logarithmically for GUE, enforcing the plateau:

$$K_{\text{plateau}} = \frac{t_H}{2\pi} = N\rho_c, \quad t > t_H. \tag{14}$$

Because the scrambled subspace has finite dimension $N \sim e^{S_{\text{th}}}$, the level spacing is $\Delta \sim e^{-S_{\text{th}}}$, giving $t_H \sim e^{S_{\text{th}}}$ as expected [2, 19].

3.5 The Disconnected SFF and the Dip

The connected SFF (13) captures the universal Wigner-Dyson correlations of the scrambled sector but does not by itself produce the dip. The full SFF decomposes as

$$K(t) = K_{\text{disconn}}(t) + K_{\text{conn}}(t), \tag{15}$$

where the disconnected piece is

$$K_{\text{disconn}}(t) = \left| \int \rho_c(\epsilon)\, e^{-i\epsilon t}\, d\epsilon \right|^2 = |\hat{\rho}_c(t)|^2, \tag{16}$$

the squared modulus of the Fourier transform of the mean compressed spectral density ρ_c.

Proposition 3.8 (Decay of the disconnected SFF)**.** *Let ρ_c be the mean spectral density of $\mathcal{H}_\alpha^{(N)}$ obtained from the subordination equation* (11)*. Since μ (the spectral measure of H_{post}) is compactly supported, the free compression theorem guarantees that ρ_c is also compactly supported and continuous on its support. Therefore, by the Riemann-Lebesgue lemma,*

$$K_{\text{disconn}}(t) = |\hat{\rho}_c(t)|^2 \xrightarrow{t\to\infty} 0. \tag{17}$$

The decay rate is set by the smoothness of ρ_c: if $\rho_c \in C^k$, then $|\hat{\rho}_c(t)| = O(t^{-k})$.

Remark 3.9 (The dip as a crossover)**.** Since $K_{\text{disconn}}(t)$ decays from a large initial value (of order $N^2\rho_c^2$ at $t = 0$) while $K_{\text{conn}}(t)$ grows linearly from zero, they must cross. The dip minimum of the full SFF occurs at the Thouless time t_{Th}, defined implicitly by

$$K_{\text{disconn}}(t_{\text{Th}}) \approx K_{\text{conn}}(t_{\text{Th}}) = \frac{t_{\text{Th}}}{2\pi}. \tag{18}$$

For $t < t_{\text{Th}}$, the disconnected piece dominates and the SFF falls. For $t_{\text{Th}} < t < t_H$, the ramp dominates and the SFF rises.

Analogy with JT gravity. This structure is the kinematic analog of the replica wormhole decomposition in JT gravity (Section 2.3). In JT gravity, the disconnected disk contributions play the role of K_{disconn}, and the double-trumpet wormhole plays the role of K_{conn}. In JT gravity, we would start with Schwarzian energy density with max energy cutoff as an input and determine the required properties of H_{post}.

Remark 3.10 (Per-instance spectral resolution and branch permanence)**.** The SFF plateau has an implication that goes beyond the universal ramp-plateau statistics. For a *fixed* instance α, the compressed Hamiltonian $\mathcal{H}_\alpha^{(N)}$ has a discrete spectrum with level spacing $\Delta \sim e^{-S_{\rm th}}$. On the onset of the plateau at $t_H \sim e^{S_{\rm th}}$, individual eigenstates of $\mathcal{H}_\alpha^{(N)}$ are spectrally resolved - the system can 'distinguish' between them. At times $t > t_H$, the state of the full system $S \cup E$ has settled into the eigenstate structure of $\mathcal{H}_\alpha^{(N)}$ within $\mathcal{V}_\alpha$, and the Poincaré recurrence time $t_{\rm rec} \sim e^{e^{S_{\rm th}}}$ ensures this resolution is permanent on all physically relevant timescales. This is the per-instance analog of what replica wormholes establish in JT gravity: for each fixed theory instance, the late-time state is pure and the spectrum is discrete. The physical consequence - that different branches of the global state $|\Psi_{SE}\rangle$ are associated with distinct, permanently separated energy eigenstates of $\mathcal{H}_\alpha^{(N)}$ - is developed further in Section 3.6 to argue that S is effectively in a definite outcome branch for each fixed α.

3.6 Late-Time Purity from the Replica Trick and Wormhole Saddles

We now establish, directly from the gravitational path integral (despite the name 'gravitational path integral', this is really 'geometrical path integral' over how replicas of the system are to be pieced together and nothing else), that the reduced state ρ_S is approximately pure at late times, though conditionally. This result is the key ingredient that permits the Born rule derivation of Sections 4–6 to operate within unitary deterministic quantum mechanics, without assuming wavefunction collapse or the Everettian interpretation. The argument follows the replica wormhole approach of [6] applied to $\mathrm{Tr}(\rho_S^n)$ and does *not* invoke the quantum extremal surface formula or the islands prescription [9].

Note that in case non-ensemble systems are discussed, the gravitational path integrals that incorporate contributions from Z_n^{conn} (which corresponds to K_{conn}) may not be justified.

Also, we could say that our kinematic ansatz provides a different context to justify replica wormhole calculations - connected Euclidean replicas (or replica topology) of system S correspond to (integrating over) unknown Haar-random subspace contributions.

An important *caveat* must be noted. Despite being theoretically appealing, it is not yet clear how far these replica wormhole calculations can be made general, beyond the duality between double-scaled random matrix theory under Schwarzian leading-order energy density and JT gravity. Therefore, our statement regarding final purity of ρ_S should be taken in this conditional context.

The replica trick. The n-th Rényi entropy of ρ_S is

$$S_n(\rho_S) = \frac{1}{1-n} \log \mathrm{Tr}(\rho_S^n), \quad n \geq 2, \tag{19}$$

and the von Neumann entropy is recovered as $S(\rho_S) = \lim_{n\to 1} S_n(\rho_S)$. In the gravitational path-integral formulation, $\mathrm{Tr}(\rho_S^n)$ is computed by preparing n identical copies (replicas) of the total system $S \cup E$ and identifying boundary conditions cyclically across replicas: the final state of replica k is glued to the initial state of replica $k+1 \pmod n$ at the S boundary, while the E boundaries are left free and summed over. The gravitational instruction is to sum over all bulk geometries consistent with these cyclic boundary conditions.

Competing saddles: disconnected disks and connected wormholes. Two classes of saddle-point geometries dominate the sum:

1. **Disconnected saddle.** The n replicas remain geometrically independent - each contributes its own disk topology, with no bulk connectivity between replicas. This gives

$$\mathrm{Tr}(\rho_S^n)\big|_{\mathrm{disconn}} = \frac{Z_n^{\mathrm{disconn}}}{Z_1^n}, \tag{20}$$

   where $Z_1 = \mathrm{Tr}(e^{-\beta H})$ is the single-replica partition function. In the high-entropy regime $S_{\mathrm{th}} \gg \log d$, this saddle evaluates to

$$S_n(\rho_S)\big|_{\mathrm{disconn}} \approx \log d, \tag{21}$$

   the maximum entropy of $\mathcal{H}_S$, consistent with the approximately maximally mixed state established in Corollary 4.4.

2. **Connected (wormhole) saddle.** All n replicas are joined by a single bulk Euclidean wormhole geometry connecting all n asymptotic boundaries. Evaluating the on-shell action of this connected topology gives

$$\mathrm{Tr}(\rho_S^n)\big|_{\mathrm{conn}} = \frac{Z_n^{\mathrm{conn}}}{Z_1^n}. \tag{22}$$

   In the fully scrambled regime ($t > t_{\mathrm{Page}}$), the wormhole geometry is preferred by the path

integral, and $Z_n^{\text{conn}} \to Z_1^n$ up to corrections exponential in $-S_{\text{th}}$, so that

$$\operatorname{Tr}(\rho_S^n)\big|_{\text{conn}} \to 1, \quad S_n(\rho_S)\big|_{\text{conn}} \to 0. \tag{23}$$

The min-rule and saddle dominance. The physical prescription, which follows directly from taking the dominant saddle in the gravitational path integral, is [6] (also see [7]):

$$S_n(\rho_S) = \min\big(S_n\big|_{\text{disconn}},\ S_n\big|_{\text{conn}}\big). \tag{24}$$

This is about which bulk topology has lower on-shell (gravitational) action, i.e., which saddle dominates the path integral. The min-rule requires only the sum over topologies inherent in the path integral and is independent of any quantum extremal surface construction. Before the Page time ($t < t_{\text{Page}}$), the disconnected saddle is preferred and $S_n \approx \log d$ (the entropy is high, as in Corollary 4.4). After the Page time ($t > t_{\text{Page}}$), the connected wormhole saddle is preferred and $S_n \to 0$ (the entropy decreases toward zero). This is precisely the descent half of the Page curve.

Proposition 3.11 (Late-Time Purity from 'Wormhole' Dominance)**.** *Under the kinematic framework, for $t_{\text{Page}} \ll t \ll t_{\text{rec}}$, the connected wormhole saddle dominates the replica path integral for $\operatorname{Tr}(\rho_S^n)$ for all $n \geq 2$. Consequently,*

$$\operatorname{Tr}(\rho_S^n) \to 1 + O\big(e^{-S_{\text{th}}}\big), \quad \forall\, n \geq 2, \tag{25}$$

which in the $n \to 1$ limit implies $S(\rho_S) \to 0$. The reduced state ρ_S is exponentially close to a pure state: there exists a unit vector $|\psi_S(\alpha)\rangle \in \mathcal{H}_S$ (depending on the instance α) such that

$$\|\rho_S - |\psi_S(\alpha)\rangle\langle\psi_S(\alpha)|\|_1 \leq \epsilon_{\text{wh}}, \tag{26}$$

where $\epsilon_{\text{wh}} = O(e^{-S_{\text{th}}/2}) \to 0$ exponentially in S_{th}.

Proof sketch. For $t > t_{\text{Page}} \sim e^{S_{\text{th}}/2}$, the on-shell action of the connected n-replica wormhole geometry is lower than the action of n disconnected disks. This follows from the fact that the gravitational free energy of the interior has decreased below the Bekenstein-Hawking entropy of the unscrambled system: in the notation of the kinematic framework, the subspace $\mathcal{V}_\alpha$ has exhausted the coarse-grained entropy of the initial state, so the connected geometry is no longer

suppressed relative to the disconnected one. In the fully scrambled regime, $Z_n^{\text{conn}}/Z_1^n \to 1$ up to corrections exponential in $-S_{\text{th}}$, by the same spectral rigidity that enforces the SFF plateau (Section 3.4): the N eigenvalues of $\mathcal{H}_\alpha^{(N)}$ are spectrally rigid on scale $\Delta \sim e^{-S_{\text{th}}}$, so ratios of partition functions converge at the same rate. Hence (25) holds. The purity bound (26) follows from the standard inequality: if $\text{Tr}(\rho_S^2) \geq 1-\epsilon$, then ρ_S has a leading eigenvalue $\lambda_{\max} \geq 1-\epsilon$, so $\|\rho_S - |\psi\rangle\langle\psi|\|_1 \leq 2\sqrt{\epsilon}$ for the corresponding eigenvector $|\psi\rangle$. □

4 Bipartite Entanglement and Page's Theorem

4.1 Bipartite Decomposition

We partition the total degrees of freedom into a local system S of dimension d and environment E such that $\mathcal{H}_{\text{total}} = \mathcal{H}_S \otimes \mathcal{H}_E$, where $\dim \mathcal{H}_E = N_E$. A generic initial state of S in the measurement basis $\{|s_i\rangle\}_{i=1}^n$ (the eigenbasis of the observable to be measured) takes the form $|\psi_S\rangle = \sum_{i=1}^n c_i|s_i\rangle$, with $\sum_i |c_i|^2 = 1$. Under the joint unitary evolution generated by H of Eq. (7), the full state $|\Psi_{SE}\rangle$ evolves into the scrambled sector, entangling S with E.

Proposition 4.1 (Branch preservation under environmental scrambling)**.** *Let $|\Psi_{SE}\rangle = \sum_k c_k|s_k\rangle|E_k\rangle$ be any bipartite pure state with $\{|s_k\rangle\}$ orthonormal in $\mathcal{H}_S$ and $\langle E_j|E_k\rangle = \delta_{jk}$. For any unitary U_E acting on $\mathcal{H}_E$ alone,*

$$(I_S \otimes U_E)|\Psi_{SE}\rangle = \sum_k c_k|s_k\rangle(U_E|E_k\rangle), \tag{27}$$

and the transformed branch states $\{U_E|E_k\rangle\}$ remain mutually orthogonal, since unitaries preserve inner products: $\langle E_j|U_E^\dagger U_E|E_k\rangle = \delta_{jk}$. Consequently the reduced state $\rho_S = \text{Tr}_E[|\Psi_{SE}\rangle\langle\Psi_{SE}|] = \sum_k |c_k|^2|s_k\rangle\langle s_k|$ is exactly invariant under all environmental scrambling, however chaotic.

4.2 Maximal Entropy via Page's Theorem

Theorem 4.2 (Page, 1993 [8])**.** *Let $|\Psi\rangle$ be a uniformly random pure state on a bipartite Hilbert space $\mathcal{H}_S \otimes \mathcal{H}_E$ with $\dim \mathcal{H}_S = d \leq \dim \mathcal{H}_E = N_E$. Then the average von Neumann entropy of the reduced density matrix $\rho_S = \text{Tr}_E|\Psi\rangle\langle\Psi|$ satisfies*

$$\langle S(\rho_S)\rangle = \ln d - \frac{d}{2N_E} + O\left(\frac{d^2}{N_E^2}\right). \tag{28}$$

In our setting, $N_E \sim e^{S_{\text{th}}}$, so $N_E \gg d$ for any macroscopic S, and the correction is exponentially suppressed:

$$\langle S(\rho_S)\rangle \xrightarrow{N_E\to\infty} \ln d \equiv S_{\max}. \tag{29}$$

Page's result is sharp: the standard deviation of $S(\rho_S)$ about its mean is $\sigma \sim d/(N_E \ln 2)$, which also vanishes exponentially.

Remark 4.3 ((no) tension with Proposition 3.11)**.** Page's theorem gives $\langle S(\rho_S)\rangle \approx \log d$ (maximal entropy), while Proposition 3.11 gives $S(\rho_S) \to 0$ (near-zero entropy). There is no contradiction, because these results operate on different objects. Page's theorem is an *ensemble* result - it describes the average entropy of ρ_S over the Haar measure on α. Proposition 3.11 is a *per-instance* typicality result - it describes the entropy of ρ_S for a fixed instance α at late times $t > t_{\text{Page}}$. The ensemble average of a quantity that is zero for each α can be non-zero only if the measure on α assigns finite weight to instances that are not yet in the post-Page regime; but the coarse-grained (Haar-averaged) description captures the entropy profile of a typical α at intermediate times (before full Page-time scrambling). In the Born rule derivation of Section 5, Page mixing (Corollary 4.4) is used to establish *equal* probabilities across fine-grained branches; purity (Proposition 3.11) is used separately to confirm that S occupies a *single* branch.

Corollary 4.4 (Kinematically Forced Maximally Mixed State)**.** *Under the kinematic constraint, the reduced density matrix ρ_S is exponentially close to the maximally mixed state:*

$$\left\|\rho_S - \frac{I_d}{d}\right\|_1 \le \sqrt{\frac{d}{N_E}} \xrightarrow{N_E\to\infty} 0. \tag{30}$$

Proof. From the purity formula implied by Page's theorem, $\mathrm{Tr}(\rho_S^2) \approx 1/d + 1/N_E$, combined with the Cauchy-Schwarz inequality $\|A\|_1 \le \sqrt{d}\|A\|_2$: $\|\rho_S - I_d/d\|_1 \le \sqrt{d}\|\rho_S - I_d/d\|_2 = \sqrt{\mathrm{Tr}(\rho_S^2) - 1/d} \approx \sqrt{1/N_E}\cdot d^{1/2} \sim \sqrt{d/N_E}$. □

4.3 Asymptotic Probability Bounds from Page Mixing

The following lemma bridges the mixing result of Corollary 4.4 and the envariance argument of Section 5.

Lemma 4.5 (Asymptotic Equiprobability)**.** *Under the kinematic constraint, for any measure-*

ment outcome described by a rank-one projector $E_k = |s_k\rangle\langle s_k|$ on $\mathcal{H}_S$:

$$\left|p_k - \frac{1}{d}\right| = \left|\mathrm{Tr}\left[E_k\left(\rho_S - \frac{I_d}{d}\right)\right]\right| \leq \|E_k\|_\infty \cdot \left\|\rho_S - \frac{I_d}{d}\right\|_1 \leq \sqrt{\frac{d}{N_E}}. \tag{31}$$

In particular, $|p_j - p_k| \leq 2\sqrt{d/N_E} \sim 2e^{-S_{\mathrm{th}}/2}$ for all j, k.

Proof. The first inequality is the duality bound $|\mathrm{Tr}[AB]| \leq \|A\|_\infty \|B\|_1$. Since E_k is a rank-one projector, $\|E_k\|_\infty = 1$. The second inequality is Corollary 4.4. □

5 Envariance and the Derivation of Equal Probabilities

5.1 Axioms of Envariance

In the strict thermodynamic limit $N_E \to \infty$, Corollary 4.4 implies that the global pure state $|\Psi_{SE}\rangle$ converges to the symmetric maximally entangled form. We work in this limit throughout this section; the finite-N_E corrections are bounded by Lemma 4.5.

Definition 5.1 (Envariance Axioms [20, 21])**.** Let $|\Psi_{SE}\rangle$ be a pure bipartite state. An operation u_S acting on $\mathcal{H}_S$ is called *envariant* if there exists a counter-operation $\tilde{u}_E$ on $\mathcal{H}_E$ such that

$$(u_S \otimes \tilde{u}_E)|\Psi_{SE}\rangle = |\Psi_{SE}\rangle. \tag{32}$$

The axioms are:

(A1) **State-dependence** (Symmetry/Structure): The probability of a measurement outcome depends only on the quantum state $|\Psi_{SE}\rangle$ and on the observable, not on the history by which the state was prepared.

(A2) **Causality / No-Signaling** (Physics): An envariant operation u_S acting on $\mathcal{H}_S$ cannot change any local observable of S. If u_S is envariant, then for any observable O_S on $\mathcal{H}_S$,

$$\mathrm{Tr}[O_S \rho_S] = \mathrm{Tr}[O_S(u_S \rho_S u_S^\dagger)]. \tag{33}$$

(A3) **Additivity** (Measure Theory): For disjoint, mutually exclusive fine-grained events, probabilities are additive.

The continuity assumption required for irrational amplitudes is stated separately as Assumption 6.1; it is an independent regularity postulate not derived from (A1)–(A3).

Axiom (A2) is the no-signaling theorem in relativistic quantum field theory [22]: operations on E cannot propagate information to S outside the light cone. The envariance condition ensures that any transformation of S can in principle be undone by remote action on E; causality then forces the local state to be invariant under such transformations. The logical roles of the axioms in the probability argument are:

- **(A1) + state symmetry** $\Rightarrow$ envariance of swap unitaries (algebra, no probability assumed).
- **(A2) / no-signaling** $\Rightarrow$ equal outcome probabilities for symmetric swaps (physics).
- **(A3) / additivity** $\Rightarrow$ Born rule for rational $|c_i|^2$ (measure theory).
- **Continuity (Assumption 6.1)** $\Rightarrow$ extension to all $|c_i|^2$ (regularity).

5.2 Proof of Equal Probabilities for Maximally Entangled States

Theorem 5.2 (Equal Probabilities from Envariance)**.** *In the thermodynamic limit $N_E \to \infty$, let $|\Psi_{SE}\rangle$ be the maximally entangled state*

$$|\Psi_{SE}\rangle = \frac{1}{\sqrt{d}} \sum_{k=1}^{d} |s_k\rangle \otimes |e_k\rangle, \tag{34}$$

with $\{|s_k\rangle\}$ and $\{|e_k\rangle\}$ orthonormal bases. Then, under Axioms (A1)–(A3)*, the probability p_k of measuring the state $|s_k\rangle$ is $p_k = 1/d$ for all k.*

Proof. Consider the swap unitary $U_S^{(jk)}$ exchanging $|s_j\rangle \leftrightarrow |s_k\rangle$. Direct computation gives:

$$\left(U_S^{(jk)} \otimes U_E^{(jk)}\right) |\Psi_{SE}\rangle = |\Psi_{SE}\rangle, \tag{35}$$

so $U_S^{(jk)}$ is envariant with counter-operation $\tilde{u}_E = U_E^{(jk)}$. This is a purely algebraic statement about the state vector; no probabilistic assumption enters.

By Axiom (A2), envariance of $U_S^{(jk)}$ implies that for any local observable O_S: $\mathrm{Tr}[O_S \rho_S] = \mathrm{Tr}[O_S (U_S^{(jk)} \rho_S U_S^{(jk)\dagger})]$. Applying this to $O_S = |s_j\rangle\langle s_j|$: the left side is p_j; the right side is the probability of outcome j in the swapped state, which is the probability of outcome k in the original state, i.e. p_k. Therefore $p_j = p_k$.

The physical content of Axiom (A2) is as follows. If $p_j \neq p_k$, then applying $U_S^{(jk)}$ would change the outcome distribution at S - a locally detectable event. But since $U_S^{(jk)}$ is envariant,

its effect could then be undone by $U_E^{(jk)}$ acting only on E. An agent with access only to S would observe the distribution change back, implying that a remote operation on E has altered local statistics at S - a violation of no-signaling. Therefore $p_j = p_k$ for all j, k.

Since the probabilities are equal and sum to unity, $p_k = 1/d$ for all k. □

Remark 5.3 (Necessity of maximal entanglement)**.** For a non-maximally entangled state $|\Phi\rangle = \sum_k \lambda_k |s_k\rangle|e_k\rangle$ with $|\lambda_j| \neq |\lambda_k|$, the swap $U_S^{(jk)}$ is not envariant: its counter-operation would require complex amplitude manipulations in E, not a simple swap. The kinematic constraint of Section 4, specifically Corollary 4.4 and Lemma 4.5, is therefore essential: it ensures the global state is exponentially close to the maximally entangled form in the thermodynamic limit.

Remark 5.4 (Basis-independence of the equal-probability result)**.** For the maximally entangled state (34), the equal-probability result $p_k = 1/d$ holds for *any* orthonormal basis $\{|a_k\rangle\}$ on $\mathcal{H}_S$, not only for the specific basis $\{|s_k\rangle\}$ written in the theorem. For any such basis, the swap $U_S^{(jk)}$ exchanging $|a_j\rangle \leftrightarrow |a_k\rangle$ is envariant: since all Schmidt coefficients are equal, any unitary on S can be countered by a corresponding unitary on E that restores the state. The no-signaling argument of the proof therefore applies in whatever basis the measurement is performed. The Born rule derivation does *not* require a pre-selected pointer basis from decoherence or any other mechanism, *if* we are assured that interaction preserves amplitude information on S from initial time to final time.

6 Coarse-Graining and the General Born Rule

6.1 Rational Approximation and Fine-Grained Branches

Theorem 5.2 gives equal probabilities only for maximally entangled states with equal Schmidt coefficients. To derive the general Born rule for arbitrary amplitudes c_i, we use the coarse-graining argument that is in inverse direction of [20, 21], adapted to our kinematic setting. Instead of fine-graining states with unequal amplitudes, we coarse-grain states of equal amplitudes onto the measurement basis/observable.

Let $|\psi_S\rangle = \sum_{i=1}^n c_i|s_i\rangle$ with $|c_i|^2 = m_i/M$, where m_i are positive integers and $M = \sum_i m_i$. (The irrational case is treated in Section 6.3.) The kinematic constraint ensures $N_E \sim e^{S_{\text{th}}} \gg M$, so the fine-grained partition is feasible.

Coarse-graining. The following step is a *Hilbert-space refinement* - a relabeling of states - not a dynamical claim, consistent with the measurement basis/observable of system S. This step is all about respecting unitarity - if S was initially on unequal amplitudes in this measurement basis, then this information has to be preserved in case unitarity is respected, even though the fine-grained state has equal amplitudes.

We introduce M orthonormal environmental states grouped into n blocks of sizes $m_1, \ldots, m_n$: $\{|e_{i,j}\rangle : j = 1, \ldots, m_i\}$ correlated with pointer state $|s_i\rangle$. The entangled state is then written as:

$$|\Psi_{SE}\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m_i} \frac{c_i}{\sqrt{m_i}} |s_i\rangle \otimes |e_{i,j}\rangle. \tag{36}$$

Given $|c_i|^2 = m_i/M$, we have

$$\frac{|c_i|}{\sqrt{m_i}} = \frac{\sqrt{m_i/M}}{\sqrt{m_i}} = \frac{1}{\sqrt{M}}, \quad \forall\, i. \tag{37}$$

This is an arithmetic identity following directly from the rational condition $|c_i|^2 = m_i/M$ - not a consequence of any dynamical property of the scrambling. The kinematic content of the framework enters in two ways: (i) the environment must have $N_E \gg M$ to accommodate the fine-grained partition, which is guaranteed since $N_E \sim e^{S_{\text{th}}}$; and (ii) in the thermodynamic limit, the Haar-random isometry generates global states with the entanglement structure (36), as a consequence of the Haar measure's unitary invariance distributing Schmidt weights uniformly across the environmental partition.

Eq. (36) therefore simplifies to:

$$|\Psi_{SE}\rangle = \frac{1}{\sqrt{M}} \sum_{\mu=1}^{M} e^{i\phi_\mu} |s_\mu\rangle \otimes |e_\mu\rangle, \tag{38}$$

where μ is a fine-grained index, $|s_\mu\rangle$ denotes the coarse-grained pointer state $|s_i\rangle$ to which the μ-th fine-grained branch belongs, and $e^{i\phi_\mu}$ are arbitrary phases that do not affect probabilities.

6.2 Application of Envariance to the Fine-Grained State

The fine-grained state (38) is exactly the maximally entangled form (34) with d replaced by M. By Theorem 5.2, every fine-grained branch is assigned equal probability:

$$p_\mu = \frac{1}{M}, \quad \forall\, \mu \in \{1, \ldots, M\}. \tag{39}$$

Summing over the m_i fine-grained branches correlated with the coarse-grained outcome $|s_i\rangle$ using Axiom (A3):

$$p_i = \sum_{j=1}^{m_i} p_{i,j} = \frac{m_i}{M} = |c_i|^2. \tag{40}$$

The Born rule for rational squared amplitudes is thereby derived.

6.3 Extension to Irrational Amplitudes

Assumption 6.1 (Continuity of the probability map)**.** The probability functional $p_i : \mathbb{S}^{2n-1} \to [0, 1]$ is continuous in the ℓ^2 norm on state space.

This is an independent regularity postulate, not derivable from Axioms (A1)–(A3). It is analogous to the measurability condition in Gleason's theorem [23], which similarly cannot be derived from the algebraic structure of the Hilbert space alone. The assumption is physically reasonable - small state perturbations should produce small measurement probability perturbations - but we list it explicitly as a postulate rather than a derived result, in order not to obscure the logical structure of the argument.

Given Assumption 6.1: the set of states with rational squared amplitudes is ℓ^2-dense in $\mathbb{S}^{2n-1}$. For any state $|\psi_S\rangle$ with irrational $|c_i|^2$ and any $\epsilon > 0$, there exists $|\psi_S^\epsilon\rangle$ with rational squared amplitudes such that $\||\psi_S\rangle - |\psi_S^\epsilon\rangle\|_2 < \epsilon$. Therefore, $p_i(|\psi_S\rangle) = \lim_{\epsilon\to 0} p_i(|\psi_S^\epsilon\rangle) = \lim_{\epsilon\to 0} |c_i^\epsilon|^2 = |c_i|^2$.

7 Conclusion

We presented a kinematic framework in which the SFF dip-ramp-plateau structure and the Born rule of quantum measurement admit a unified reconstruction. This was done by treating the late-time scrambled sector as an effective Haar-random subspace - motivated by the ETH, maximum entropy, and holographic considerations.

When it is applicable to associate replica wormhole calculations to the dip-ramp-plateau SFF results, main local system S has a final pure state for each subspace instance α. This allows us to avoid the measurement problem and stay within deterministic unitary quantum mechanics for the derivation of the Born rule. Probability in this case is classical, associated with epistemic and ensemble ignorance of exact α. Once α is known, then probability disappears.

The conditions for replica wormhole calculations are expected to be somewhat restrictive, and quantum interpretations (such as collapse or Everettian interpretations) may be assumed

instead. We sacrifice deterministic quantum mechanics, but the Haar-random ansatz still provides a much stronger argument for the Born rule.

Consider the Everettian interpretation. Interpreting a quantum state as a superposition of different branches, with only one branch being observable, does not directly provide the Born rule even after the preferred basis problem is resolved. That unitary quantum mechanics alone, with few auxiliary postulates (and axioms), provides the Born rule in the Haar-random ansatz case provides a much stronger evidence for the derived nature of the Born rule.

One of the criticisms of the envariance argument for the Born rule is that fine-graining a state with unequal amplitudes to a state with equal amplitudes is problematic, as it implicitly assumes that every fine-grained state in the universe has effectively equal amplitudes - easily broken by unitary transformation. By reversing the order - first presenting a context where a *fine-grained* state with effectively equal amplitudes is reached from an initial system state (of S) with unequal amplitudes and then coarse-graining to achieve unequal amplitudes - we circumvented this criticism, placing the envariance argument in a stronger position.

This paper does have one important limitation - while largely inspired by quantum gravity, the kinematic Haar-random ansatz is simply assumed rather than derived. We leave the explicit derivation to future works.

Data availability and declaration of interests

The author(s) have no funding source to declare. There is no conflict of interests.

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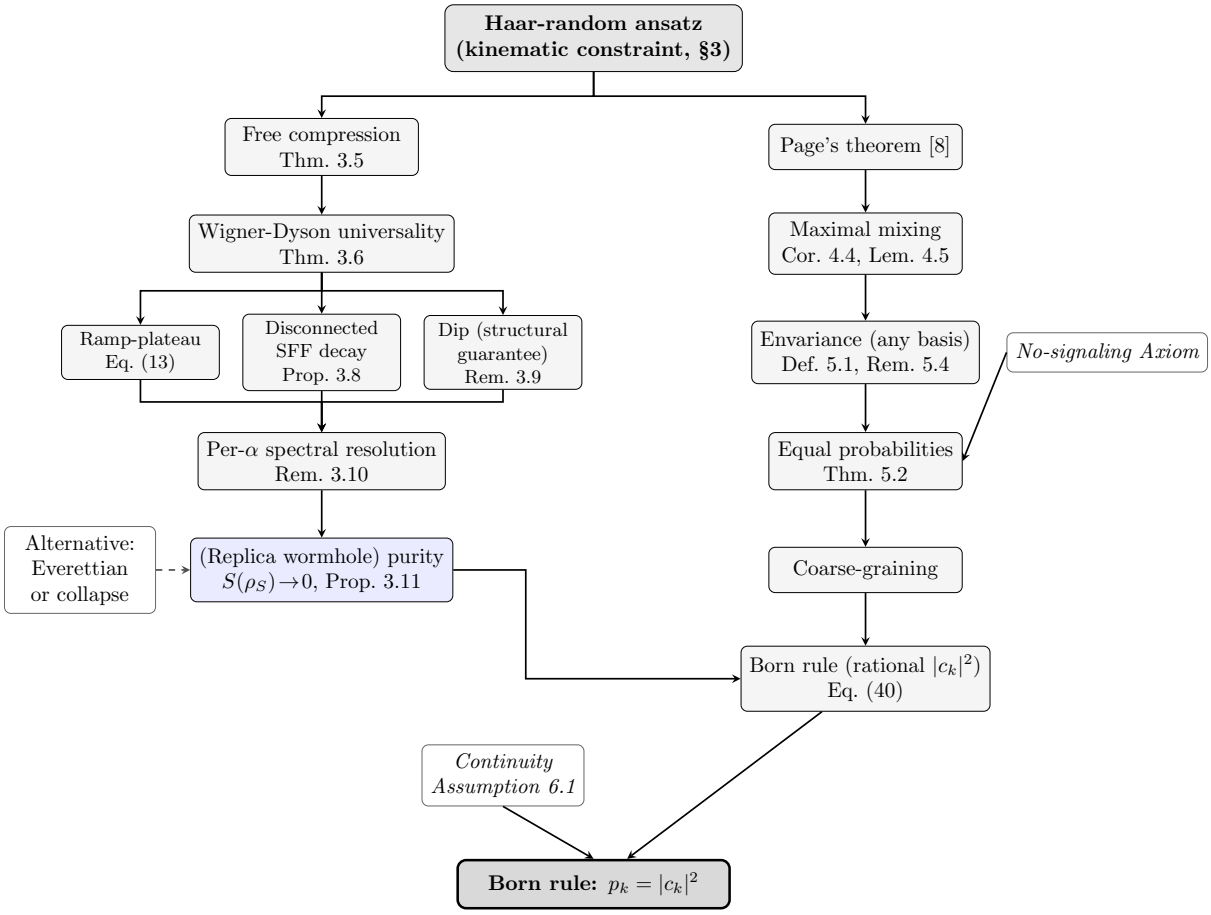

Figure 1: Logical structure of the paper. Wigner-Dyson/SFF results are largely restatements of existing works, though the dip-ramp-plateau SFF structure only arises for specific setups - most conveniently JT gravity. A small contribution of this paper in this regard is replacing theory randomness with subspace randomness associated with α, leading to effectively the same SFF results. Blue box is Proposition 3.11: the replica trick applied to the gravitational path integral, summing disconnected disk vs. connected wormhole saddles, establishes typicality result $S(\rho_S) \to 0$ at late times without the islands formula. This purity result supplies the "in the language of JT gravity replica wormholes, black holes evaporate unitarily" justification that enables the Born rule derivation within unitary deterministic quantum mechanics. The dashed arrow from the "Alternative" box indicates that Everettian or collapse assumptions can substitute for Proposition 3.11 if desired. The rest of the paper is about the Page theorem justification of the envariance derivation of the Born rule, which relies on the same Haar-random ansatz - though the envariance argument involves coarse-graining instead of usual fine-graining.